RCS03 | Instructor Guide

CTS Cohort 6 | CTS Day 11 | 2026-07-22 | 10:00 AM-3:00 PM | 6 students

Electrical Fundamentals: Receptacles and Box Layout

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Big idea: Residential wiring is a controlled path; professionals de-energize, verify, route, terminate, inspect, and document.

Architecture of the day

Time	Instructor move
10:00-10:15	Arrival, PPE, retrieval warm-up
10:15-10:30	Morning quiz
10:30-10:50	Quiz review and energy-boundary check
10:50-11:20	Hot, neutral, and grounding path model
11:20-12:00	Box-layout and receptacle termination demonstration
12:00-12:30	Verification ladder guided practice and lab release
12:30-1:00	Lunch and controlled reset
1:00-2:25	Two-box, two-receptacle de-energized capstone lab
2:25-2:45	Visual, tug, continuity, evidence, cleanup
2:45-3:00	Exit ticket and Day 4 preview

Learning targets

- Trace hot, neutral, and equipment-grounding paths in a training circuit
- Lay out two receptacle boxes on the capstone wall
- Prepare and terminate conductors on training receptacles
- Complete visual, mechanical, and continuity checks while de-energized

Materials and setup

- Six isolated training wall stations; no premises-power connection
- Two device boxes and two training receptacles per student
- Training cable/conductors, grounding hardware, labels, and approved diagrams
- Strippers, hand tools, and manufacturer-directed termination tools
- Meters reserved for de-energized continuity checks
- Eye protection, drilling controls, and evidence devices

Instructor talking points

- Retrieve PV101 circuit-path and meter habits without assuming field authorization.
- Separate conductor purpose from color shorthand; verify the training diagram.
- Demonstrate box layout, cable protection, jacket entry, conductor length, stripping, clockwise terminal loops, torque-by-manufacturer, and grounding.
- Teach the inspection ladder: de-energized verification, visual check, tug test, continuity check, instructor approval.
- Explain that Oregon-regulated installations require appropriate permits, inspections, and licensed work.

Whiteboard plan

1. Source -> protection -> switch/load -> return
2. Hot / neutral / ground
3. De-energize -> lock/tag -> verify
4. Two receptacle boxes
5. Visual -> mechanical -> meter -> document

Lab facilitation

Students mount two training device boxes in their capstone walls, route a de-energized training cable path, prepare conductors, terminate two receptacles, and complete a four-part verification. No connection to building power is permitted.

Lab sequence

6. Energy-control briefing
7. Verify training source isolated
8. Lay out two boxes
9. Instructor layout approval
10. Mount boxes
11. Route training cable
12. Prepare conductors
13. Terminate receptacles
14. Visual and tug inspection
15. De-energized continuity check
16. Label, photograph, reset

Safety boundaries

- Training mock-up only. Keep every circuit de-energized, isolated from premises wiring, and under instructor control. This class does not authorize unlicensed field electrical work.
- Treat any unverified conductor as energized

- Eye protection during drilling and cable preparation
- No damaged insulation or exposed copper outside terminals
- Continuity mode only on isolated circuits
- Instructor controls any test source

Evidence targets

- Circuit-path worksheet
- Box-layout measurements
- Two completed receptacle terminations
- Visual/tug/continuity log
- Instructor approval record

Quiz answer key

1. What conductor normally carries current from the source to the load? The ungrounded or hot conductor.

2. What conductor normally provides the return path? The grounded or neutral conductor.

3. What is the purpose of the equipment-grounding conductor? It provides a low-impedance fault-current path and bonds equipment; it is not a normal return path.

4. What must be proven before continuity mode is used? The circuit is de-energized and isolated.

5. What is a tug test checking? That a termination is mechanically secure.

6. Why must copper not extend loosely outside a terminal? It can contact other parts and create a fault or shock hazard.

7. May this mock-up be connected to building power? No.

8. Name the four verification stages used today. De-energized verification, visual check, mechanical/tug check, and continuity/meter check.

9. Who approves the box layout before drilling? The instructor.

10. Does course practice authorize unlicensed field electrical work? No.

Exit-ticket look-fors

- Draw the training circuit path
- Name one strong termination feature
- State the meter safety boundary
- Name one inspection you completed

Google Classroom post

RCS Day 3 - Electrical Fundamentals: Receptacles and Box Layout

Today we focused on residential wiring is a controlled path; professionals de-energize, verify, route, terminate, inspect, and document. Students completed guided instruction, hands-on capstone work, verification, evidence capture, and cleanup. Submit your worksheet, lab evidence, and exit ticket before leaving.

Source and safety references

- **Canonical RCS skeleton:** /CANONICAL/00_INBOX/Downloads/rcs_curriculum_plan.md
- **Cohort 6 RCS01 source:** /CANONICAL/02_PROJECTS/CTS_Cohort_06/03_DAILY/D09_RCS01_2026-07-20/Source/lesson.md
- **OSHA PPE:** https://www.osha.gov/personal-protective-equipment/construction
- **OSHA eye and face protection:** https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.102
- **OSHA circuit lockout/tagging:** https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.417
- **Oregon electrical codes:** https://www.oregon.gov/bcd/codes-stand/pages/electrical.aspx
- **Oregon permits:** https://www.oregon.gov/bcd/lbdd/pages/oregon-permits.aspx

RCS03 | Morning Quiz - Answer Key

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Electrical Fundamentals: Receptacles and Box Layout

1. What conductor normally carries current from the source to the load?

The ungrounded or hot conductor.

2. What conductor normally provides the return path?

The grounded or neutral conductor.

3. What is the purpose of the equipment-grounding conductor?

It provides a low-impedance fault-current path and bonds equipment; it is not a normal return path.

4. What must be proven before continuity mode is used?

The circuit is de-energized and isolated.

5. What is a tug test checking?

That a termination is mechanically secure.

6. Why must copper not extend loosely outside a terminal?

It can contact other parts and create a fault or shock hazard.

7. May this mock-up be connected to building power?

No.

8. Name the four verification stages used today.

De-energized verification, visual check, mechanical/tug check, and continuity/meter check.

9. Who approves the box layout before drilling?

The instructor.

10. Does course practice authorize unlicensed field electrical work?

No.

Scoring guidance

One point per item. Accept technically equivalent wording. Use errors to reteach before lab release. Safety-boundary errors must be corrected explicitly before the student begins the related task.

RCS03 | Student Packet

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Big idea: Residential wiring is a controlled path; professionals de-energize, verify, route, terminate, inspect, and document.

Learning targets

- Trace hot, neutral, and equipment-grounding paths in a training circuit
- Lay out two receptacle boxes on the capstone wall
- Prepare and terminate conductors on training receptacles
- Complete visual, mechanical, and continuity checks while de-energized

Today's run

10:00-10:15 | Arrival, PPE, retrieval warm-up

10:15-10:30 | Morning quiz

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2:45-3:00 | Exit ticket and Day 4 preview

Key words

hot | neutral | equipment grounding conductor | receptacle | device box | sheathing | insulation | terminal | pigtail | continuity

Guided worksheet

1. Trace the training circuit path with arrows.

_____

_____

2. Label hot, neutral, and grounding conductors.

_____

_____

3. Record both box layout measurements.

_____

_____

4. List four termination defects to inspect.

_____

_____

5. Record continuity results only after the instructor confirms de-energization.

_____

_____

Afternoon lab mission

Students mount two training device boxes in their capstone walls, route a de-energized training cable path, prepare conductors, terminate two receptacles, and complete a four-part verification. No connection to building power is permitted.

Safety boundary

- Training mock-up only. Keep every circuit de-energized, isolated from premises wiring, and under instructor control. This class does not authorize unlicensed field electrical work.
- Treat any unverified conductor as energized
- Eye protection during drilling and cable preparation
- No damaged insulation or exposed copper outside terminals
- Continuity mode only on isolated circuits
- Instructor controls any test source

Build / test sequence

[ ] Energy-control briefing

[ ] Verify training source isolated

[ ] Lay out two boxes

[ ] Instructor layout approval

[ ] Mount boxes

[ ] Route training cable

[ ] Prepare conductors

[ ] Terminate receptacles

[ ] Visual and tug inspection

[ ] De-energized continuity check

[ ] Label, photograph, reset

Evidence record

[ ] Circuit-path worksheet: _____

[ ] Box-layout measurements: _____

[ ] Two completed receptacle terminations: _____

[ ] Visual/tug/continuity log: _____

[ ] Instructor approval record: _____

Reflection and exit ticket

Draw the training circuit path

_____

_____

Name one strong termination feature

_____

_____

State the meter safety boundary

_____

_____

Name one inspection you completed

_____

_____

Instructor verification

Status: [ ] Approved [ ] Correct and recheck [ ] Needs follow-up

Instructor initials: _____ Date/time: _____

RCS03 | Morning Quiz

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Name: _____ Date: _____

Answer from your own understanding. This is a teaching check, not a trap.

1. What conductor normally carries current from the source to the load?

_____

_____

2. What conductor normally provides the return path?

_____

_____

3. What is the purpose of the equipment-grounding conductor?

_____

_____

4. What must be proven before continuity mode is used?

_____

_____

5. What is a tug test checking?

_____

_____

6. Why must copper not extend loosely outside a terminal?

_____

_____

7. May this mock-up be connected to building power?

_____

_____

8. Name the four verification stages used today.

_____

_____

9. Who approves the box layout before drilling?

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10. Does course practice authorize unlicensed field electrical work?

_____

_____